

Motivation

Traditional AI vision models heavily rely on cloud computing. However, in practical applications such as autonomous driving and industrial defect detection, transmitting high-resolution images back to the cloud encounters fatal risks, including network latency, exorbitant bandwidth costs, and data privacy leaks. Shifting AI inference to the source of data generation—namely, "Edge AI"—is the ultimate solution to these problems.

Literature Study

Mainstream compression technologies include Network Pruning, Model Quantization, and Knowledge Distillation.

- Accuracy Collapse due to Outliers: When simply applying Post-Training Quantization (PTQ) to convert models to INT8, the "Activation Outliers" in the deep layers of modern high-accuracy models can cause the quantization algorithm to fail, leading to a precipitous drop in accuracy.
- Limitations of a Single Technology: Over-reliance on a single technique (such as extreme pruning) will hit "Occam's Hill," resulting in the collapse of the model's feature representation capabilities.

Technical Approach

- Main challenges: How to scale down the memory footprint and maximize hardware throughput without significantly sacrificing the prediction accuracy of YOLOv8/v11.

Rather than relying on a single technique, we are building a "Composite Compression Pipeline":

- Feature-Based Knowledge Distillation (Vanilla KD): To preserve accuracy while reducing model size, we adopt a teacher–student distillation framework tailored for object detection models such as YOLO. A large pretrained YOLO model serves as the Teacher, and a lightweight YOLO variant serves as the Student. In addition to training the student using standard detection losses (classification, objectness, and bounding box regression), we introduce a feature imitation loss between intermediate

feature maps of the teacher and student networks. Specifically, feature maps are extracted from selected backbone and neck layers of both the teacher and student networks to capture intermediate semantic representations. When the channel dimensions between the two networks differ, a 1×1 convolutional layer is introduced to project the student feature maps into the same dimensional space as the teacher. An L2 loss is then applied to minimize the discrepancy between the aligned feature representations, enabling the student model to progressively approximate the internal structural knowledge learned by the teacher. If time permits, we may further enhance the distillation stage by incorporating Localization Distillation (LD) to improve spatial supervision and bounding box localization accuracy.

- **INT8 Entropy Calibration PTQ & Granularity Optimization:** To maximize the feature representation capability after quantization, we adopt a mixed granularity strategy. For weights, we utilize Per-Channel Quantization. This is because relying on a single scale for the whole weight tensor (Per-Tensor Quantization) leads to accuracy drops in smaller models. This common failure results from large differences (more than 100x) in the ranges of weights across different output channels. We deliberately avoid Group Quantization, as it is primarily designed to balance accuracy and hardware efficiency for extremely low bit-widths (e.g., FP4 AI on Blackwell GPUs) and adds unnecessary complexity for our INT8 edge deployment. For activations, leveraging TensorRT for calibration, we minimize the loss of information by minimizing the KL divergence to find the optimal clipping threshold. This effectively resolves the accuracy degradation caused by activation outliers.
- **Kernel Fusion and Mixed Precision:** Delving into the underlying OS process, we execute kernel fusion to reduce memory transactions and ensure the inference process deeply invokes underlying hardware acceleration units (e.g., the dedicated Tensor Cores on the Jetson platform). For incompatible layers, we retain FP16 (Mixed Precision) to eliminate hardware-software friction and achieve measurable, significant

speedups. Finally, through Pareto Analysis, we will plot the Accuracy vs. Latency trade-off curve. The chart will systematically compare the performance points of three model variants: the original Baseline model, the Distilled FP32 model, and the final Quantized INT8 model, thereby providing the optimal deployment strategy.

Project Timelines

This project is expected to be completed within 8 weeks, divided into four core phases:

- Week 1-2 (Mar 9 - Mar 22) | Environment Setup and Baseline Testing: Set up the TensorRT/ONNX runtime environment on Jetson/RPi, deploy the original YOLO model, and benchmark the baseline metrics of the underlying inference process, including Latency, Throughput (FPS), Peak Memory Footprint, and Model Accuracy. (Milestone 1: Deliver the Baseline Evaluation Report)
- Week 3-4 (Mar 23 - Apr 5) | Knowledge Distillation Implementation: Establish the YOLO Teacher-Student architecture, complete the hybrid distillation training, and restore the floating-point accuracy of the lightweight model. (Milestone 2: Produce high-accuracy FP32 Student model weights)
- Week 5-6 (Apr 6 - Apr 19) | INT8 Quantization and Hardware Optimization: Execute TensorRT entropy calibration and INT8 conversion, and perform layer-wise profiling to configure mixed precision. (Milestone 3: Export an INT8 engine with no performance regression)
- Week 7-8 (Apr 20 - May 3) | System Validation and Pareto Analysis: Conduct stress tests on edge devices and plot the Accuracy vs. Latency trade-off curve. (Milestone 4: Complete the Pareto Frontier analysis and finalize the project report)

References

- Han, S. et al. (2015). Deep compression: Compressing deep neural networks with pruning, trained quantization and Huffman coding. ICLR.

- Chen, G. et al. (2017). Learning efficient object detection models with knowledge distillation. NeurIPS.
- Zheng, Z. et al. (2022). Localization distillation for dense object detection. CVPR.
- Migacz, S. (2017). 8-bit Inference with TensorRT.
- Nagel, M. et al. (2019). Data-Free Quantization Through Weight Equalization and Bias Correction. ICCV.
- Krishnamoorthi, R. (2018). Quantizing Deep Convolutional Networks for Efficient Inference: A Whitepaper. arXiv.